Leader-Based Coalition Formation for Extremely Large Scale Collectives

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***Abstract*—Robotic collectives deployed in challenging environments require efficient coordination algorithms. Coalition formation partitions robots into task-oriented teams enabling coordinated execution of complex multiple robot tasks. The coalition formation problem's computational complexity is often directly related to the collective's size and number of tasks. LeaderGRAPE-S, a hedonic game-based coalition formation algorithm was developed for extremely large scale heterogeneous robot collectives (i.e., 10000 robots) and a fixed number of tasks. These collectives' heterogeneity is restricted to a small number of robot types relative to the collective size. The developed algorithm extends a hedonic game-based coalition formation algorithm by integrating a leader-follower model to leverage the reduced heterogeneity and improve runtime efficiency. A centralized evaluation with simulated heterogeneous robot collectives of up to 10000 robots demonstrated that the LeaderGRAPE-S algorithm achieved a significant runtime reduction with a negligible increase in communication, while generating optimal solutions, making it the first algorithm to demonstrate hedonic game-based coalition formation for extremely large scale heterogeneous collectives.**

***Index Terms*—Coalition Formation, Swarms, Hedonic Games.**

I. Introduction

Robotic collectives are applicable to complex missions (e.g., military [1], disaster response [2], underwater inspection and monitoring [3]). Collectives are inspired by biological swarms (e.g., schools of fish, flocks of birds) that exhibit emergent self-organizing behaviors resulting from localized interactions [4]. Marine salps are a biological swarm [5]. Salps exist as long chains in different configurations (see Figure 1(a)) that can be up to tens to hundreds of meters in length [6]. The individual salps' simple morphology (Figure 1(b)) limits their basic functionality (e.g., feeding, locomotion) [5]. Novel soft reconfigurable underwater robots are being developed based on salps. The Landsalp prototype (shown in Figure 1(c)) replicates biological salp's multi-jet propulsion locomotion. These simplistic salp-inspired robots can individually provide at most one to two services or capabilities. Coordination in robotic collectives composed of these robots can be achieved by designing collective coalition formation algorithms.

Robotic collectives can leverage their members' diverse capabilities to accomplish complex tasks that are difficult for single or multiple robot systems to execute. Coalition formation partitions robotic collectives into task-oriented groups that combine capabilities to meet the tasks' requirements. Finding an optimal coalition structure is NP-hard [7]. The computational complexity can increase with increased collective size or services per robot or number of tasks, necessitating development of computationally efficient coalition formation methods. Hedonic game-based coalition formation provides a viable near real-time (e.g., $<$ 5 minutes) solution for heterogeneous collectives of up to 500 robots with distributed computation, and up to 1000 robots with centralized computation [8], [9], thus showing potential for extremely large scale collectives.

The presented work was supported by ONR grant N00014-23-1-2171. The views, opinions, and findings expressed are those of the authors and are not to be interpreted as representing the official views or policies of ONR.

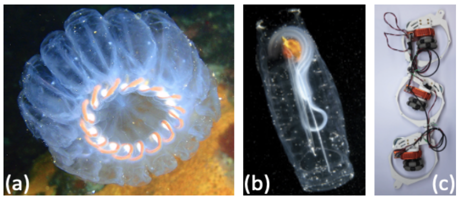

Fig. 1. (a) An individual salp zooid [10]. (b) A whorl-shaped salp chain [11]. (c) A prototype of a LandSalp robot [12].

LeaderGRAPE-S, a hedonic game-based coalition formation algorithm was developed for extremely large scale salp-inspired soft robot collectives. This algorithm leverages the collective's heterogeneity using a leader-follower approach, a concise type-based representation for robots and tasks, and the hedonic game properties to improve runtime efficiency. A simulation-based evaluation with up to 10000 robots resulted in high quality solutions with a significant runtime reduction.

II. Related Work

Collective coalition formation algorithms draw inspiration from their multiple robot variants and biological collectives [8], [13]. Coalition formation problem solutions use greedy heuristics [14], market-based auctions [15], optimization techniques [16], and game-theoretic approaches [17]. Hedonic game-based algorithms incorporate agent preferences to quickly generate Nash stable coalitions [18], [19]. An evaluation for a 1000 robot collective found hedonic games [9], [19] significantly reduced runtime without compromising solution quality [8], as compared to auctions [20], [21], making them a promising coalition formation approach for extremely large

2025 IEEE Conference on Artificial Intelligence (CAI) | 979-8-3315-2400-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/CAI64502.2025.00126

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DOI 10.1109/CAI64502.2025.00126

scale collectives. Hedonic coalition formation games' self-interested agents seek to maximize their individual utilities [22], [23]. A centralized hedonic game-based algorithm using weighted bipartite graph matching was developed [18], [24]. Distributed solutions required less communication [25], [26]. The Group Agent Partitioning and Placing Event (GRAPE) framework modeled the coalition formation problem as an anonymous hedonic game where individual agents preferred coalitions with a smaller number of co-working agents [19] and obtained Nash stable partitions in polynomial time. The algorithm's strongly connected communication network enabled asynchronous operation. The GRAPE-S solution for heterogeneous multiple robot systems uses a services model to produce near-optimal solutions in near real-time, with a 1000 robot collective [9]. Robot's high-level behaviors were modeled as *services* (e.g., mapping, navigation), based on robot characteristics and resources [13]. This manuscript uses the services model [13] for extremely large scale biologically inspired heterogeneous collectives.

A leader-follower strategy controlled a multiple robot formation by using a formation matrix to allocate robots to specific positions using auctions [27]. Real-time multiple robot formation control combined the leader-follower strategy with mobile ad-hoc communication protocol to enable mitigating issues (e.g., lost followers, disjoint formations) [28]. A resource constrained coalition formation approach's leader robots provided key resources for tasks [29]. Each leader's preliminary coalitions added followers that satisfied the task requirements, and the coalitions with maximum utility were assigned the tasks with a $O(n^3)$ computational complexity, but this approach was limited to ten robots and three tasks. These approaches' outcomes suggest suitability for extremely large scale collectives; although none of these approaches were validated for heterogeneous collectives $\geq$ 1000 robots. The LeaderGRAPE-S algorithm leverages the leader-follower hierarchy to significantly reduce the computational complexity in extremely large scale heterogeneous robot collectives.

Concise game representations can reduce computational complexity and improve scalability [30]–[32]. The notion of agent types suggested that agents of the same types had the same marginal contributions [33]. A concise characteristic function representation, facilitated by a constant number of agent types reduced computationally intractable problems with a large number of agents to polynomially tractable problems. [31]. This type-based representation reduced the characteristic function's dimensionality and computational complexity to $O(n^{2t})$ for n agents and t agent types. This approach is applied to this manuscript's extremely large collectives.

III. Problem Formulation

The coalition formation for task allocation problem optimally assigns agents to task-oriented coalitions [14]. A solution to a coalition formation problem consists of a set of *n* robots, $\Phi = \{\phi_1, ..., \phi_n\}$ and a set of *m* tasks, $T = \{t_1, ..., t_m\}$, with a coalition-task mapping $V \rightarrow \Phi_v|, \Phi_v \subseteq \Phi$, resulting in a set of *m* coalition-task pairs, $C = \{c_1, c_2, ..., c_m\} \mid c_i = \langle \Phi_i, t_i \rangle$, $\forall i \in [1, m]$. The successful task completion earns the members the task's utility. Several robot capability models (e.g., resources [13], services [13], schema-based [34]) can determine whether a coalition's combined capabilities satisfy a task's requirements. The services model is used, where each robot performs high-level behaviors or *services* (e.g., manipulation, data processing, perception) [13].

Long salp chains of individual zooids with highly simplified morphology and limited functionality inspire collective robot coalitions. Each robot in a collective is assumed to provide at most two services from a set of services, $S = \{s_1, s_2, ..., s_p\}$. Each robot ϕ_i has a service vector $S_{\phi_i} = (s_1,...,s_k) \mid k \in [1, p]$, $|S_{\phi_i}| \leq 2$. Each task $t_j's$ service vector $S_{t_j} = (s_1,...,s_k) \mid k \in [1, p]$ denotes the number of services of each type required by the task. The task's utility, u_j, is awarded if the coalition has the required services. Limiting robot heterogeneity to two services per robot can enhance the coalition formation efficiency for extremely large heterogeneous collectives. LeaderGRAPE-S leverages GRAPE-S's speed and hedonic properties, and adopts a concise representation to reduce the computational complexity by exploiting the leader-follower structure.

The LeaderGRAPE-S algorithm's hedonic game-based approach utilizes individual robot's preferences to identify coalitions. This game formulation is defined as a pair (n, $\succsim$), where n is the finite set of robots, and $\succsim$ represents each robots' preference profile over coalitions $i \in n$ [22]. Robot ϕ_i's preference for coalition c_1 over coalition c_2 is denoted as $c_1 \succsim_i c_2$. Nash stable coalition structures can be obtained when no robot has incentive to deviate from its existing coalition [22]. Robot preferences in anonymous hedonic games depend on the coalition size [19]. GRAPE-S expressed robot preferences using a peaked reward of both the robot's provided service k, and the task t_j. Thus, individual robot *i's* reward R_i for performing a service *k* in task $t_j's$ coalition c_j is:

$$R_i(t_j, c_j) = \frac{u_{jk}}{|s_k|} \times e^{-\frac{|c_{jk}|}{|s_k|}+1}, \tag{1}$$

where $|c_{jk}|$ is the number of robots assigned to perform service *k* in coalition c_j for task t_j , $|s_k|$ is the total number of robots required by task t_j to perform service *k*.

GRAPE-S had $O(n^2|S|md_G)$ computation and $O(n^3 d_G)$ communication complexity, with d_G being the communication network's diameter [8] ($d_G = 1$ for fully connected network). GRAPE-S's complexity increased with number of robots, exponentially increasing the computational and communication requirements. LeaderGRAPE-S's type-based representation and leader-follower architecture addresses this issue.

Computationally intractable coalition structure generation problems can be solved in polynomial time by limiting the number of possible robot types [30]–[32]. The coalition formation type-based representation categorizes each robot by a *type* [31], [33], where robots of the same type provide the same marginal contribution to the task. LeaderGRAPE-S's type-based representation equated a robot's service with the robot's type. A robot type is a vector of length equal to the total number of services *p*, with each element denoting the number of services of the k^{th} category provided by that robot. Accordingly, robot *i's* type ϕ_i^t for service s_3 in a problem with ten total services is $\phi_i^t = \langle 0,0,1,0,0,0,0,0,0,0 \rangle$. Similarly coalition *j's* type c_j^t for task t_j requiring two units of s_3, and five units of s_7 is represented as $c_j^t = \langle 0,0,2,0,0,0,5,0,0,0 \rangle$. This type-based robot and task-coalition representation limits the number of tasks robots must consider during coalition formation and also facilitates leaders' follower selection.

A. LeaderGRAPE-S Algorithm

The LeaderGRAPE-S algorithm combines the hierarchical leader-follower structure with GRAPE-S's speed to improve the coalition formation computational complexity for extremely large scale collectives. Leader robots equivalent to the number of services are added to the collective's robots. Each leader robot manages a single service, to which robots providing that service are assigned as followers. Robots providing multiple services are randomly assigned to one of their services' leaders. Thus, a GRAPE-S problem consisting of 1000 robots providing ten different services is equivalent to a LeaderGRAPE-S problem with 1010 robots (i.e., 1000 robots, plus ten leader robots, one for each service). The leaders conduct negotiations for task assignments among their followers, and do not actively participate in task coalitions.

Leaders maintain a list of tasks requiring their services, which leverages the concise type-based representation by only considering the leaders' and tasks' types. A task requiring multiple different services is represented in multiple leaders' task lists. Each leader runs GRAPE-S, which allows leaders to conduct independent negotiations with their followers sequentially. Each follower maintains a belief about the number of robots assigned to a task. Similar to GRAPE-S, on each iteration, each follower robot selects a preferred task-coalition using Eq. 1, broadcasts its beliefs about task assignments of all other followers associated with that leader, and updates its beliefs according to the received messages to reach consensus on a valid coalition assignment. GRAPE-S requires each of the collective's robots to iterate over all the tasks, whereas LeaderGRAPE-S's follower robots only iterate over their leader's task list. The followers negotiate until a Nash stable partition is obtained. Robots are assigned to task coalitions once all leader-follower groups complete negotiations.

1) Analysis: GRAPE-S's Nash stable coalition structure's computational complexity assumes a fully connected communication network. LeaderGRAPE-S runs GRAPE-S l times with f robots, where l is the number of leaders and f is the number of followers per leader; hence, $l = |S|$. The number of followers per leader, $f = \frac{n}{|S|}$, for an ideal case with uniform service distribution among all the collective's robots. The use of the leader-follower hierarchy and the type-based representation enables the followers to negotiate only their task assignments. This reduced negotiation allows LeaderGRAPE-S to reach Nash stability in $O(\frac{n^2 m}{|S|})$ time.

GRAPE-S has $O(n^3)$ communication complexity. LeaderGRAPE-S's communication is proportional to the total number of robots (including leaders), similar to GRAPE-S. Thus, LeaderGRAPE-S requires $O(n^3)$ communication.

TABLE I
INDEPENDENT VARIABLES FOR COMPARISON EVALUATION.

Base Collective Size	1000, 5000, 10000
Percent Tasks	1, 10, 50
Service Types: Per Robot	10:1, 10:2

IV. ALGORITHMIC COMPARISON EXPERIMENT

GRAPE-S produced near-optimal solutions (i.e., $> 95\%$ utility) in near real-time (i.e., < 5 minutes) for 1000 robot collectives [8]. Auction-based approaches [20], [21] scaled poorly and inconsistently in terms of runtime, solution optimality, and communication requirements as the collective size increased [8], making GRAPE-S a suitable baseline comparison to LeaderGRAPE-S. The hypothesis was that LeaderGRAPE-S will achieve shorter runtimes than GRAPE-S.

A. Experimental Design

Both algorithms were compared using a centralized simulation-based experiment. The independent variables (see Table I) were determined based on the design of simulated salp-inspired robot collectives. Salps' simple morphology and limited functional diversity inspired a robot design with at most two services per robot, and a collective with ten total service types, which influenced the average coalition size and the collective size. The number of tasks was expressed as a percentage of the base collective size, or *percent tasks*.

Twenty-five achievable mission problems (i.e., the collective had sufficient robots to execute all tasks) were randomly generated for each independent variable combination, resulting in a total of 450 trials per algorithm. The number of tasks, task requirements, task utilities, and individual robot services were identical for both algorithms' problems. LeaderGRAPE-S problems included leader robots equivalent to the number of service types in addition to base collective size that were not assigned to any coalition. Robots' and tasks' services were randomly selected from the total number of available service types. Task utilities were randomly assigned in the range [1, 50]. The GRAPE-S experiments were performed on a HP Z640 Workstation (Intel Xeon processor, 62 GB RAM), and the LeaderGRAPE-S experiments were performed on a Dell Precision 5820 Tower (Intel Xeon processor, 32 GB RAM). Both experiments used a centralized C++ simulator [35] with a fully-connected communication topology and instantaneous communication. Note that, both algorithms are decentralized; however, they were evaluated in a centralized fashion.

The runtime dependent variable represents the total time required to produce a coalition formation solution, and is expressed as days, hours, minutes, and seconds (DD:HH:MM:SS). Percent utility is the ratio of the solution's utility to the total possible mission utility. The total communication metric aggregates all message sizes, and is measured in megabytes (MB). Solutions with lower runtimes, higher percent utilities, and lower communication are preferred.

B. Results

LeaderGRAPE-S and GRAPE-S produced solutions for all trials. The results were analyzed using a Kruskal-Wallis test followed by Mann-Whitney-Wilcoxon post-hoc tests for statistically significant results for collective size and percent tasks. The Kruskal-Wallis effect size η^2 is based on the H-statistic. Mann-Whitney U tests analyzed performance with respect to services per robot, with the η^2 effect size based on the U-statistic. Large effect sizes ($\eta^2 > 0.14$) signify stronger influence of the particular variable on the results. The algorithms were compared using the Wilcoxon Signed-Rank test across all collective sizes, percent tasks, and services per robot, with the z-statistic representing the magnitude and direction of the difference between the compared dependent variables and the effect size reported using the Rank Biserial Correlation r_{rb}. $|r_{rb}| \approx 0.5$ indicated large effect size.

1) Runtime Results: The descriptive statistics for all collective combinations with ten service types are reported in Table II. LeaderGRAPE-S's runtime was significantly lower than GRAPE-S for all trials. LeaderGRAPE-S's longest runtime was 08:51:35 for a 10000 robot collective, 50% tasks, and two services per robot, whereas GRAPE-S's maximum runtime was 07:14:06:00. LeaderGRAPE-S's runtime increased significantly with increased collective size ($H(n = 450) = 359.69, p < 0.001, \eta^2 = 0.8$) and percent tasks ($H(n = 450) = 64.05, p < 0.001, \eta^2 = 0.13$). Similar results exist for GRAPE-S by collective size ($H(n = 450) = 355.75, p < 0.001, \eta^2 = 0.79$) and percent tasks ($H(n = 450) = 69.15, p < 0.001, \eta^2 = 0.15$). The large effect sizes indicated that collective size strongly and consistently influenced runtime. Both algorithms demonstrated significant post-hoc analyses ($p < 0.05$), indicating that runtime varied significantly between all collective sizes and between all percent tasks. LeaderGRAPE-S's and GRAPE-S's runtimes differed significantly from one to two services per robot ($p < 0.001$), with both the algorithms' runtimes being shorter for ten service types and one service per robot than with ten service types and two services per robot (LeaderGRAPE-S: ($H(n = 225) = 19375.0, p < 0.001, \eta^2 = 0.38$), GRAPE-S: ($H(n = 225) = 19963.0, p < 0.001, \eta^2 = 0.39$)). Experiments with two services per robot exhibited a larger runtime variance, due to convergence requiring more iterations.

LeaderGRAPE-S's and GRAPE-S's runtimes differed significantly across all collective sizes, percent tasks and services per robot values ($z(n = 450) = -18.38, p < 0.001, r_{rb} = 1.0$). The effect size of 1 indicated that LeaderGRAPE-S substantially and consistently outperformed GRAPE-S.

2) Utility Results: LeaderGRAPE-S generated 100% utility solutions for all trials. GRAPE-S produced optimal solutions for all collective sizes and percent tasks with ten service types and one service per robot. GRAPE-S produced near optimal solutions (i.e., utility $\geq 95\%$) for all two services per robot combinations of all collective sizes, percent tasks and ten service types. The smallest utility, 95.6%, occurred for 1000 robots with two services per robot, and 1% tasks.

TABLE II
THE RUNTIME (DD:HH:MM:SS) DESCRIPTIVE STATISTICS [MEAN (STD)] FOR TEN SERVICE TYPES BY ALGORITHM, COLLECTIVE SIZE, NUMBER OF TASKS, AND NUMBER OF SERVICES PER ROBOT.

Collective Size	GRAPE-S	LeaderGRAPE-S
1 Service Per Robot		
1% tasks		
1000	:13 (:00)	:00 (:00)
5000	:27:08 (:20)	:01:38 (:01)
10000	:03:36:36 (:01:32)	:14:01 (:05)
10% tasks		
1000	:59 (:00)	:05 (:00)
5000	:02:06:00 (:54)	:10:36 (:05)
10000	:17:00:00 (:05:55)	:01:28:48 (:38)
50% tasks		
1000	:02:15 (:01)	:07 (:00)
5000	:04:52:12 (:02:20)	:15:43 (:06)
10000	01:14:22:48 (:33:26)	:02:12:00 (:54)
2 Services Per Robot		
1% tasks		
1000	:23 (:05)	:02 (:00)
5000	:45:50 (:10:36)	:06:52 (:02)
10000	:06:02:24 (:01:03:36)	:56:44 (:20)
10% tasks		
1000	:02:40 (:37)	:02 (:00)
5000	:05:28:12 (:53:43)	:42:45 (:19)
10000	01:19:00:00 (:09:46:48)	:05:53:24 (:02:55)
50% tasks		
1000	:05:43 (:01:31)	:28 (:00)
5000	:12:18:00 (:02:37:48)	:01:03:00 (:29)
10000	04:11:12:00 (01:03:36:00)	:08:44:24 (:02:29)

3) Communication Results: LeaderGRAPE-S's communication requirement varied across collective sizes and services per robot, but was not effected by percent tasks, as shown in Figure 2. The minimum communication for LeaderGRAPE-S was 24.21 MB for a 1000 robot collective with 1% tasks, ten service types and one service per robot, while the maximum, 4.8 GB, occurred for 10000 robots with 50% tasks, ten service types and two services per robot. LeaderGRAPE-S's communication for 1000, 5000, and 10000 robot collectives was 24.21 MB, 601.07 MB, and 2402.15 MB, respectively. LeaderGRAPE-S's communication nearly doubled with two services per robot compared to one service per robot ($U(n = 225) = 16875.0, p < 0.001, \eta^2 = 0.33$). The required communication increased significantly with increased collective size for both one and two services per robot irrespective of the percent tasks ($H(n = 450) = 410.51, p < 0.001, \eta^2 = 0.91$).

GRAPE-S's communication requirement varied across collective size for ten service types and one service per robot, whereas the communication requirement varied across collective size, percent tasks, and services per robot for ten service types and two services per robot, as shown in Figure 2. GRAPE-S's minimum communication, 24 MB, occurred for 1000 robots with 1% tasks, ten service types and one service per robot, while the maximum, 9.7 GB, occurred with a 10000 robot collective for 50% tasks, ten service types and two services per robot. GRAPE-S's communication for 1000, 5000, and 10000 robot collectives was 24.00 MB, 600.00 MB, and 2400.00 MB, respectively. GRAPE-S's communication requirements increased significantly with increased collective size ($H(n = 450) = 405.12, p < 0.001, \eta^2 = 0.90$),

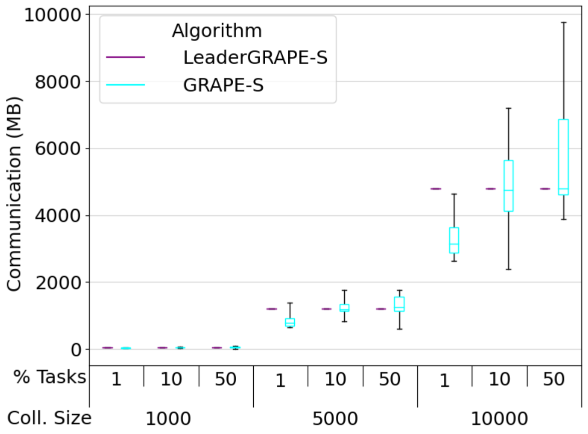

Fig. 2. Communication by collective size and percent tasks for ten service types and two services per robot.

but not by percent tasks. GRAPE-S's communication varied significantly from one to two services per robot $(U(n = 225) = 17137.5, p < 0.001), \eta^2 = 0.34)$.

The communication requirements between the algorithms differed significantly across all collective sizes, percent tasks, and services per robot combinations $(z(n = 450) = -9.54, p < 0.001, r_{rb} = 0.76)$. The Wilcoxon Signed-Rank test indicated that LeaderGRAPE-S required more communication than GRAPE-S in 379 out of 450 trials.

C. Discussion

LeaderGRAPE-S's runtimes were faster and exhibited lower variability for all independent variable combination trials. This runtime reduction is due to a) the type-based representation allowing leaders to select a subset of tasks for allocation based on their services, and b) the leader-follower hierarchy facilitating negotiations among a smaller subset of robots (i.e., each leader's followers) for a reduced number of tasks (i.e., a subset of the total tasks chosen by each leader). This distinction became more pronounced in extremely large scale collectives (i.e., 10000 robots), where LeaderGRAPE-S produced solutions in minutes, while GRAPE-S required several hours to days. LeaderGRAPE-S's reduced runtime can enable faster pre-mission coalition formation, and potentially shorten the time required for coalition reformation in response to dynamic events, thereby reducing the overall mission duration. Both algorithms' runtimes increased with the collective size and the number of tasks. LeaderGRAPE-S benefitted from conducting negotiations within smaller subsets of robots (i.e., followers), and generated Nash stable partitions more quickly.

All LeaderGRAPE-S trials produced optimal solutions. A higher solution quality can boost the applicability of both algorithms to domains that necessitate long-term and near-optimal performance (e.g., surveillance, search and rescue, disaster response). Both algorithms' communication increased with increased collective size. LeaderGRAPE-S's higher communication requirement was proportional to the number of additional robots (i.e., leaders). Leaders independently coordinated negotiations among their respective followers by sequentially running GRAPE-S in their subgroups. The number of leaders was equivalent to the number of services, and the problem modeling assumed the number of services to be significantly smaller than the collective size to replicate real world collectives. Thus, the cost of the LeaderGRAPE-S's communication overhead was negligible compared to the significant runtime improvement obtained over GRAPE-S. However, both algorithms' high communication requirements still pose a major limitation to deploying extremely large scale collectives in real world environments. Both algorithms require a fully connected communication network and assume instantaneous message passing, both of which are difficult to guarantee in real world scenarios. Developing methods to reduce communication, while still maintaining LeaderGRAPE-S's runtime efficiency is necessary to extend its applicability to uncertain and dynamic environments.

V. Service Types Evaluation

This evaluation compared LeaderGRAPE-S's performance for two different numbers of service types. The hypothesis was that more service types will result in shorter runtimes.

TABLE III
Independent variables for Service Types Variation.

Percent Tasks	1, 10
Service Types: Per Robot	10:1, 20:1

A. Experimental Design

A 10000 robot collective with one service per robot was used. The percent tasks were either 1% or 10%, while the number of service types was ten or twenty (see Table III). The number of leader robots corresponded to the number of service types, and the leaders were added to the collective (e.g., 20 service types, 20 leaders + 10000 robots = 10020 robots). Twenty-five achievable mission problems were generated for each combination, resulting in a total of 100 trials. The dependent variables, the computational hardware, and the simulation setup were identical to Section IV-A. LeaderGRAPE-S was compared for ten and twenty service types using Mann-Whitney U tests, with effect size measured using η^2.

B. Results

1) Runtime Results: The mean runtimes for 10 services were longer (1% tasks: 14:01 [:04], 10% tasks: 01:28:48 [:37]) than the 20 service results (1% tasks: 08:04 [:01], 10% tasks: 28:59 [:13]). Increasing the number of services significantly reduced the runtime $(H(n = 100) = 1875.0, p < 0.001, \eta^2 = 0.75)$. The large effect size indicated that the number of service types had a strong influence on the runtime.

2) Utility Results: LeaderGRAPE-S produced optimal solutions (i.e., 100% utility) for all experiments with twenty service types for both percent task values.

3) Communication Results: LeaderGRAPE-S's communication was 2.4 GB for all trials indicating that increasing the number of service types did not impact communication.

C. Discussion

LeaderGRAPE-S's runtime decreased with increased service types, consistent with its $O(\frac{n^2 m}{|S|})$ complexity, thereby supporting the experiment's hypothesis. The number of leaders increased with an increase in the number of services based on the assumption that each leader represented one service. An increase in number of leaders for the same collective size resulted in fewer followers assigned to each leader, which allowed leaders to more quickly reach Nash stability, thereby reducing the overall runtime. Overall, LeaderGRAPE-S maintained optimal solutions and did not require increased communication, demonstrating its adaptability towards handling diverse number of service types.

VI. Conclusion

Robotic collectives deployed for complex tasks in challenging environments require efficient coordination algorithms. LeaderGRAPE-S, a hedonic game-based coalition formation algorithm, supports allocating tasks to extremely large scale heterogeneous robot collectives. The algorithm's reduced computational complexity and solution optimality for 10000 robot collectives demonstrates its effectiveness in leveraging heterogeneity for collective coalition formation.

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